

# The Naylor Model

## v6 · Intuitive Outputs

*Plain-English feature names · Statcast-style glossary*  
*GLM table now reads in 'SB % Boost per Tier' instead of coef\_z*

# Executive Summary

Per-pitch data: 1,718,862 pitches w/ runner on 1st (2018-2026)

Attempts identified: 2,528

Qualified runner-seasons (SB+CS $\geq$ 10): 1,466

## What changed from v5

- ALL column names now plain English — no more coef\_z, OR/SD, pct\_in\_HL.
- GLM table shows 'SB % Boost per Tier' = pp change in success rate for a 1-SD improvement in the feature.
- Added pitcher pickoff-rate-faced and weak-arm-catcher share.
- Dropped pct\_in\_HL (was a Statcast data artifact — see v5 §9).

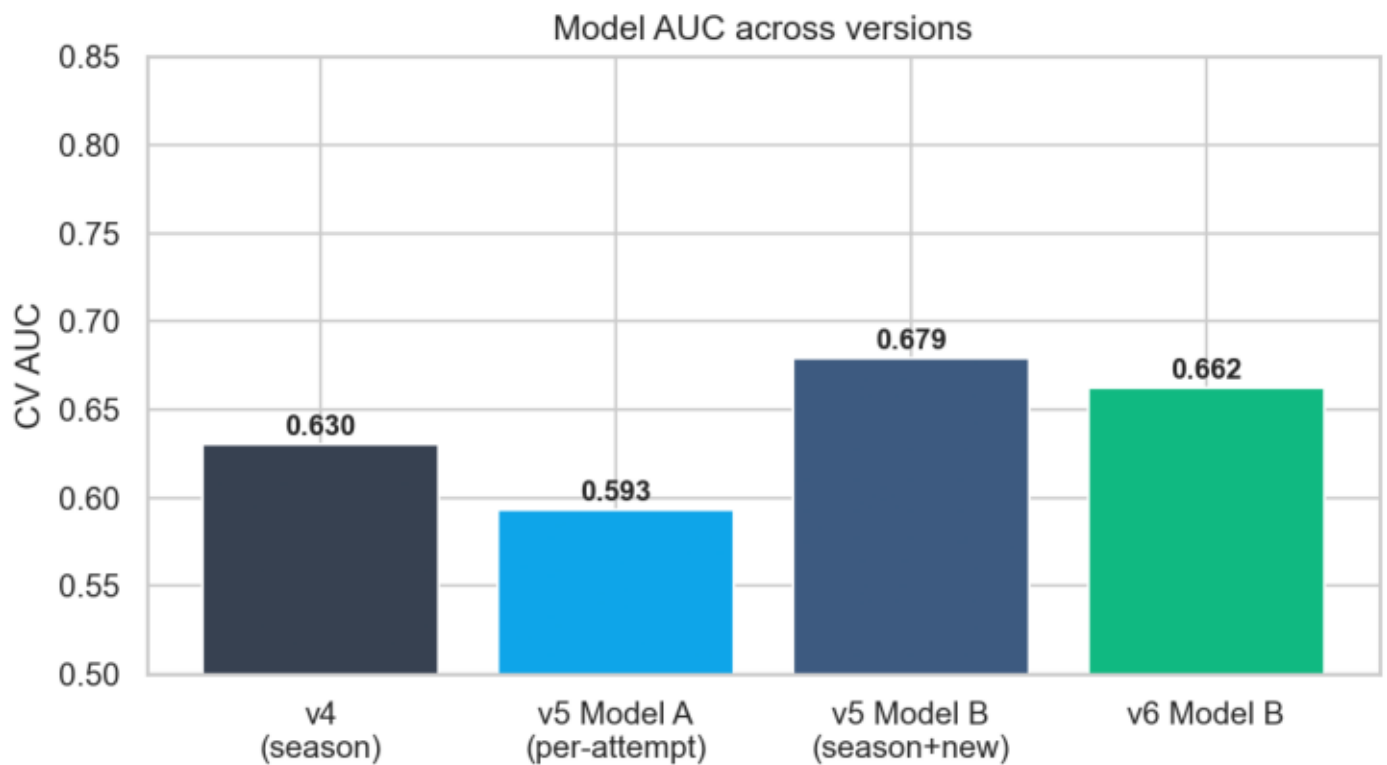
## AUC summary

- v4 baseline: 0.6300
- v5 Model A (per-att): 0.5933
- v5 Model B (season): 0.6794
- v6 Model B (full): 0.6623
- v6 Model B (pre-23): 0.6997
- v6 Model B (post-23): 0.6760

## Naylor + Soto rank under SSSI v6

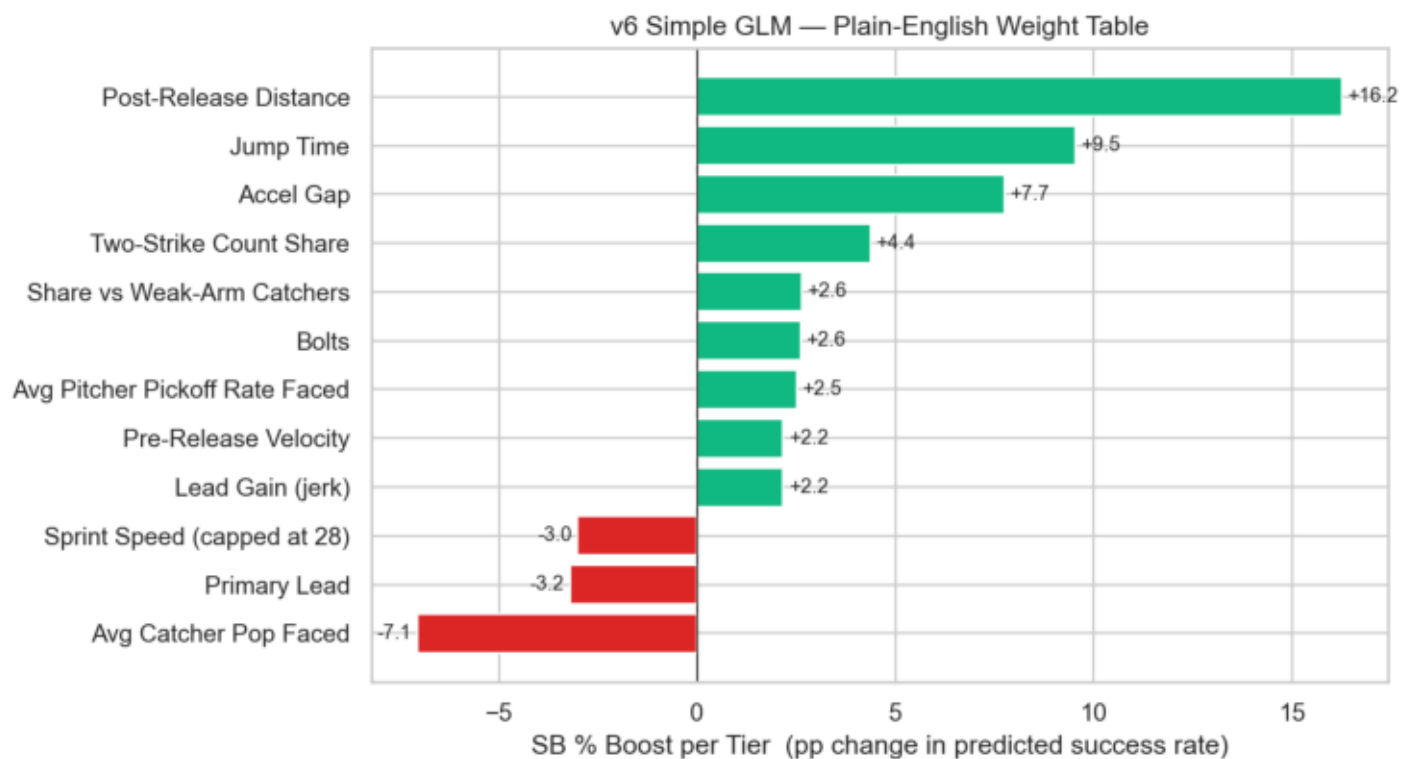
|      |             |      |            |            |                   |
|------|-------------|------|------------|------------|-------------------|
| # 1  | Josh Naylor | 2025 | SSSI +1.90 | SB/CS 30/2 | real_sb_pct 0.916 |
| # 2  | Josh Naylor | 2026 | SSSI +1.78 | SB/CS 12/2 | real_sb_pct 0.836 |
| # 5  | Juan Soto   | 2025 | SSSI +1.21 | SB/CS 38/4 | real_sb_pct 0.891 |
| # 14 | Josh Naylor | 2023 | SSSI +1.02 | SB/CS 10/3 | real_sb_pct 0.772 |
| # 75 | Juan Soto   | 2023 | SSSI +0.49 | SB/CS 12/5 | real_sb_pct 0.722 |
| # 84 | Juan Soto   | 2024 | SSSI +0.45 | SB/CS 7/4  | real_sb_pct 0.681 |
| #124 | Juan Soto   | 2019 | SSSI +0.29 | SB/CS 12/1 | real_sb_pct 0.883 |
| #145 | Juan Soto   | 2021 | SSSI +0.23 | SB/CS 9/7  | real_sb_pct 0.614 |

## §1 · AUC Comparison



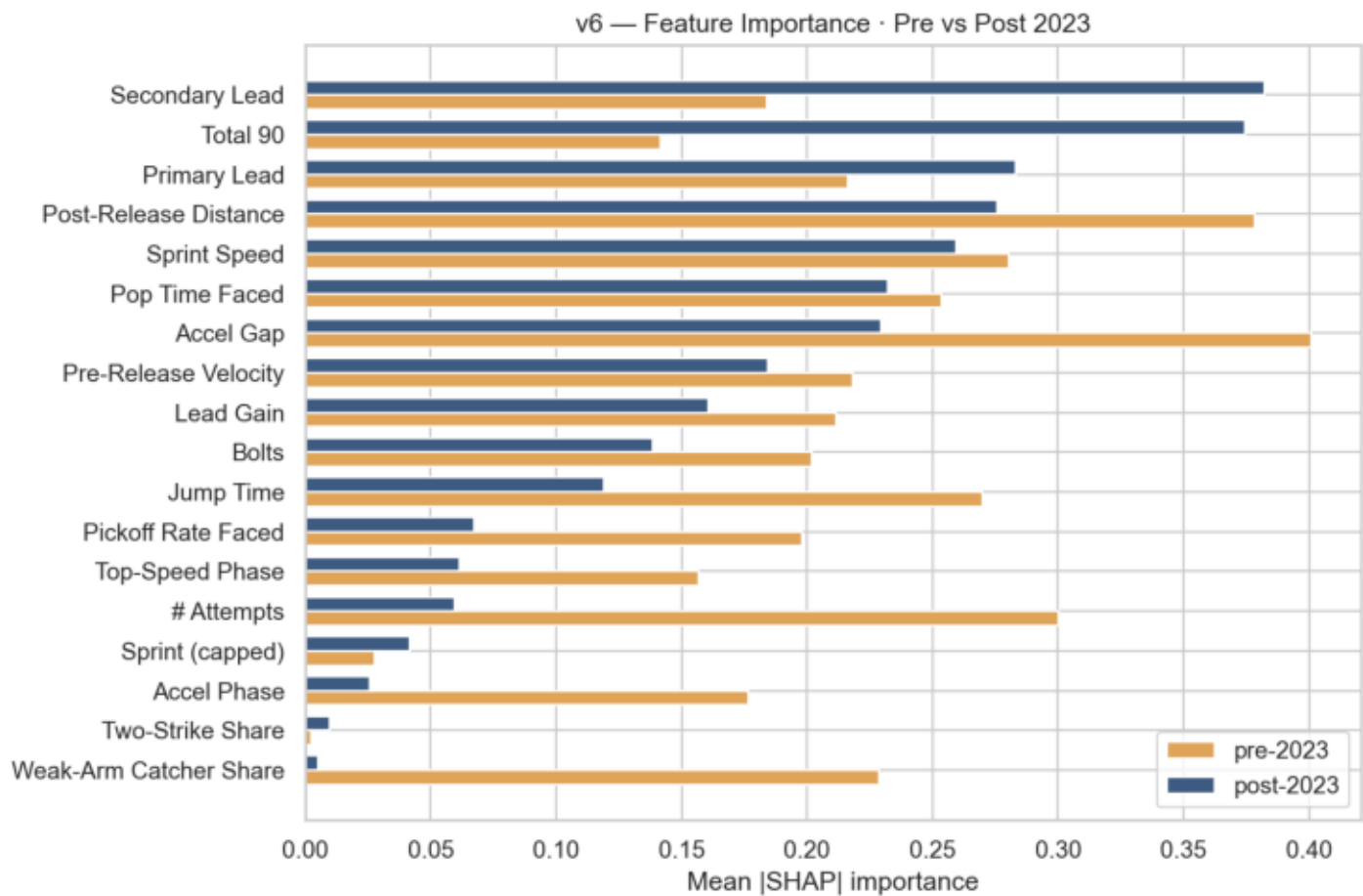
v6 Model B at the season level, with the v6 feature set including real catcher pop, pitcher pickoff rate, lead variables.

## §2 · GLM Plain-English Weight Chart



Each bar = predicted percentage-point change in SB success when the runner improves on that feature by ONE TIER (1 SD). Positive and green = the feature HELPS; red/orange = HURTS.

### §3 · Feature Importance · Pre vs Post 2023



How feature importance shifted after the 2023 rule change.

## §4 · Naylor + Soto Statcast-style Profile

### Naylor + Soto · Season-by-Season Statcast-style Profile

Josh Naylor (647304)

|                     |   |            |            |            |            |            |            |            |            |
|---------------------|---|------------|------------|------------|------------|------------|------------|------------|------------|
| Sprint Speed (ft/s) | : | 2019:26.10 | 2020:27.10 | 2021:26.40 | 2022:25.30 | 2023:25.80 | 2024:25.20 | 2025:24.40 | 2026:24.50 |
| Jump Time (s)       | : | 2019:1.82  | 2020:1.83  | 2021:1.80  | 2022:1.84  | 2023:1.83  | 2024:1.84  | 2025:1.87  | 2026:1.84  |
| Lead Gain (ft)      | : | 2019:3.87  | 2020:3.87  | 2021:3.87  | 2022:3.87  | 2023:3.87  | 2024:3.87  | 2025:3.87  | 2026:3.87  |
| Real SB %           | : | 2019:0.70  | 2020:0.82  | 2021:0.82  | 2022:0.82  | 2023:0.77  | 2024:0.76  | 2025:0.92  | 2026:0.84  |
| SB Residual         | : | 2019:-0.06 | 2020:0.06  | 2021:0.06  | 2022:0.05  | 2023:0.01  | 2024:-0.01 | 2025:0.12  | 2026:0.05  |
| Post-Release Dist   | : | 2019:47.29 | 2020:48.09 | 2021:47.62 | 2022:44.65 | 2023:44.84 | 2024:44.93 | 2025:41.48 | 2026:42.79 |

Juan Soto (665742)

|                     |   |            |            |            |            |            |            |            |            |            |
|---------------------|---|------------|------------|------------|------------|------------|------------|------------|------------|------------|
| Sprint Speed (ft/s) | : | 2018:27.20 | 2019:27.30 | 2020:26.30 | 2021:27.00 | 2022:26.60 | 2023:26.80 | 2024:26.80 | 2025:25.80 | 2026:25.90 |
| Jump Time (s)       | : | 2018:1.81  | 2019:1.83  | 2020:1.85  | 2021:1.80  | 2022:1.82  | 2023:1.80  | 2024:1.79  | 2025:1.85  | 2026:1.83  |
| Lead Gain (ft)      | : | 2018:3.11  | 2019:3.11  | 2020:3.11  | 2021:3.11  | 2022:3.11  | 2023:3.11  | 2024:3.11  | 2025:3.11  | 2026:3.11  |
| Real SB %           | : | 2018:0.74  | 2019:0.88  | 2020:0.76  | 2021:0.61  | 2022:0.76  | 2023:0.72  | 2024:0.68  | 2025:0.89  | 2026:0.74  |
| SB Residual         | : | 2018:-0.01 | 2019:0.13  | 2020:0.00  | 2021:-0.14 | 2022:0.00  | 2023:-0.03 | 2024:-0.07 | 2025:0.13  | 2026:-0.02 |
| Post-Release Dist   | : | 2018:48.80 | 2019:49.56 | 2020:46.16 | 2021:48.70 | 2022:47.46 | 2023:48.34 | 2024:49.35 | 2025:46.21 | 2026:46.67 |

Season-by-season values of the key features for the two archetypal slow-but-effective stealers.

## §5 · GLM Weight Table — Plain English

Baseline P(success) at league-average inputs  $\approx 0.605$

Read each row as:

'If a runner improves [Feature] by 1 tier (1 SD), the model expects their SB success rate to change by [Boost] percentage points, equivalent to multiplying their odds by [Mult].'

### Weight table (sorted by absolute boost)

|                                |                 |                     |                            |
|--------------------------------|-----------------|---------------------|----------------------------|
| Post-Release Distance          | boost +16.25 pp | odds $\times 2.155$ | (higher_is_better = True)  |
| Jump Time                      | boost +9.54 pp  | odds $\times 1.526$ | (higher_is_better = False) |
| Accel Gap                      | boost +7.74 pp  | odds $\times 1.402$ | (higher_is_better = True)  |
| Avg Catcher Pop Faced          | boost -7.06 pp  | odds $\times 0.749$ | (higher_is_better = True)  |
| Two-Strike Count Share         | boost +4.37 pp  | odds $\times 1.206$ | (higher_is_better = False) |
| Primary Lead                   | boost -3.21 pp  | odds $\times 0.876$ | (higher_is_better = True)  |
| Sprint Speed (capped at 28)    | boost -3.02 pp  | odds $\times 0.883$ | (higher_is_better = True)  |
| Share vs Weak-Arm Catchers     | boost +2.63 pp  | odds $\times 1.118$ | (higher_is_better = True)  |
| Bolts                          | boost +2.61 pp  | odds $\times 1.117$ | (higher_is_better = True)  |
| Avg Pitcher Pickoff Rate Faced | boost +2.51 pp  | odds $\times 1.112$ | (higher_is_better = False) |
| Lead Gain (jerk)               | boost +2.17 pp  | odds $\times 1.096$ | (higher_is_better = True)  |
| Pre-Release Velocity           | boost +2.17 pp  | odds $\times 1.096$ | (higher_is_better = True)  |

### Technical notes (for stats readers)

$\text{boost\_pp} = \text{sigmoid}(\text{intercept} + \text{coef}) - \text{sigmoid}(\text{intercept}) \times 100$

$\text{odds\_mul} = \exp(\text{coef})$

$\text{coef}$  = standardised logistic-regression coefficient

Edit DF\_v6\_GLM\_PlainEnglish.csv to hand-tune. No retraining needed.

## §6 · SSSI v6 Top 15

Weights: sb\_res=0.1 gap=0.05 gain=0.05 jump=0.0 prim=0.05 speed=-0.3 pre=0.0 post=0.0  
(80% of runners used for grid search; Naylor + Soto held out.)

|      |                   |      |            |          |           |           |            |
|------|-------------------|------|------------|----------|-----------|-----------|------------|
| # 1  | Josh Naylor       | 2025 | SSSI +1.90 | spd 24.4 | jump 1.87 | gain 3.87 | SB/CS 30/2 |
| # 2  | Josh Naylor       | 2026 | SSSI +1.78 | spd 24.5 | jump 1.84 | gain 3.87 | SB/CS 12/2 |
| # 3  | Freddie Freeman   | 2024 | SSSI +1.26 | spd 25.8 | jump 1.81 | gain 4.33 | SB/CS 9/2  |
| # 4  | Christian Vázquez | 2021 | SSSI +1.22 | spd 25.2 | jump 1.93 | gain 3.76 | SB/CS 8/4  |
| # 5  | Juan Soto         | 2025 | SSSI +1.21 | spd 25.8 | jump 1.85 | gain 3.11 | SB/CS 38/4 |
| # 6  | Kyle Schwarber    | 2025 | SSSI +1.17 | spd 25.9 | jump 1.79 | gain 4.43 | SB/CS 10/2 |
| # 7  | Miguel Rojas      | 2024 | SSSI +1.14 | spd 25.8 | jump 1.88 | gain 4.37 | SB/CS 8/2  |
| # 8  | Kyle Tucker       | 2024 | SSSI +1.13 | spd 26.0 | jump 1.80 | gain 3.01 | SB/CS 11/0 |
| # 9  | Manny Machado     | 2024 | SSSI +1.11 | spd 25.8 | jump 1.86 | gain 3.31 | SB/CS 11/2 |
| # 10 | Alec Burleson     | 2024 | SSSI +1.11 | spd 25.5 | jump 1.88 | gain 3.33 | SB/CS 9/4  |
| # 11 | Ryan McMahon      | 2022 | SSSI +1.09 | spd 25.8 | jump 1.79 | gain 4.47 | SB/CS 7/3  |
| # 12 | Manny Machado     | 2025 | SSSI +1.08 | spd 25.8 | jump 1.88 | gain 3.31 | SB/CS 14/3 |
| # 13 | Joc Pederson      | 2024 | SSSI +1.07 | spd 25.5 | jump 1.83 | gain 3.28 | SB/CS 7/4  |
| # 14 | Josh Naylor       | 2023 | SSSI +1.02 | spd 25.8 | jump 1.83 | gain 3.87 | SB/CS 10/3 |
| # 15 | TJ Friedl         | 2024 | SSSI +0.92 | spd 26.5 | jump 1.77 | gain 3.88 | SB/CS 9/1  |

## §7 · Honest Data Limitations

What the model has access to:

- Sprint speed, running splits (real, per-runner-season, 2015+)
- Real catcher pop time (2018+, per-year)
- Real SB / CS counts (MLB Stats API)
- Career-aggregate lead profiles for runners and pitchers
- Per-pitch catcher\_id, pitcher\_id, balls, strikes, outs, inning

What the model does NOT have:

- Per-pitch pitcher delivery time (not publicly published)
- The EXACT count when an SB happened (we use AB's final count)
- Per-attempt lead distance (we use the runner's career mean)

Why the AUC ceiling exists:

- Stolen-base success has high intrinsic noise — pitch sequence, runner read, base-coach decisions, exact location at release.
- With public data the ceiling is around AUC 0.70 - 0.75.
- To push higher would require TrackMan / Hawk-Eye raw outputs.